US Solar Resource Data

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Contents

Introduction

- Data source: NLR API - Solar Resource data - solar data types for given US locations: `<https://developer.nlr.gov/docs/solar/solar-resource-v1/>`
- Output fields of interest: avg_dni, avg_lat_tilt, avg_ghi
- Locations: New York, Los Angeles, Austin
- Goal: extract solar resource data for three US cities in JSON format, convert to a combined dataframe, and visualise monthly averages by location
- Research question:
  - if I install a 4kW rooftop solar panel in each city, how much energy would it generate per month?
  - how many months per year does each city exceed a viable solar threshold?
  - how much money is saved by installing the rooftop solar panel in each city?

Project Set-Up

```r
library(tidyverse)
library(httr)
library(jsonlite)
library(dotenv)
library(kableExtra)
library(formatR)
load_dot_env(".env")
knitr::opts_chunk$set(tidy = TRUE, tidy.opts = list(width.cutoff = 60))
base_url <- "https://developer.nlr.gov"

# define locations
locations <- list(
  list(name = "New York",    lat = 41,    lon = -75),
  list(name = "Los Angeles", lat = 34.05, lon = -118.24),
  list(name = "Austin",      lat = 30.27, lon = -97.74)
)

theme_set(theme_minimal())
```

API Function

```r
# reusable function to query the solar resource API and
# return a tidy dataframe
solar_api_json_get_df <- function(endpoint, queries = list()) {
    url <- paste0(base_url, "/", endpoint)
    response <- GET(url, query = queries)
    if (http_error(response)) {
        print(status_code(response))
        print(http_status(response))
        stop("Something went wrong.", call. = FALSE)
    }
    if (http_type(response) != "application/json") {
        stop("API did not return json", call. = FALSE)
    }
    json_text <- content(response, "text")
    json_lists <- jsonlite::fromJSON(json_text)
    outputs <- json_lists$outputs
    df <- outputs[c("avg_dni", "avg_ghi", "avg_lat_tilt")] %>%
        lapply(function(x) unlist(x$monthly)) %>%
        as.data.frame() %>%
        `rownames<-`(NULL) %>%
        mutate(month = month.abb) %>%
        select(month, everything())
    df
}
```

Querying All Locations

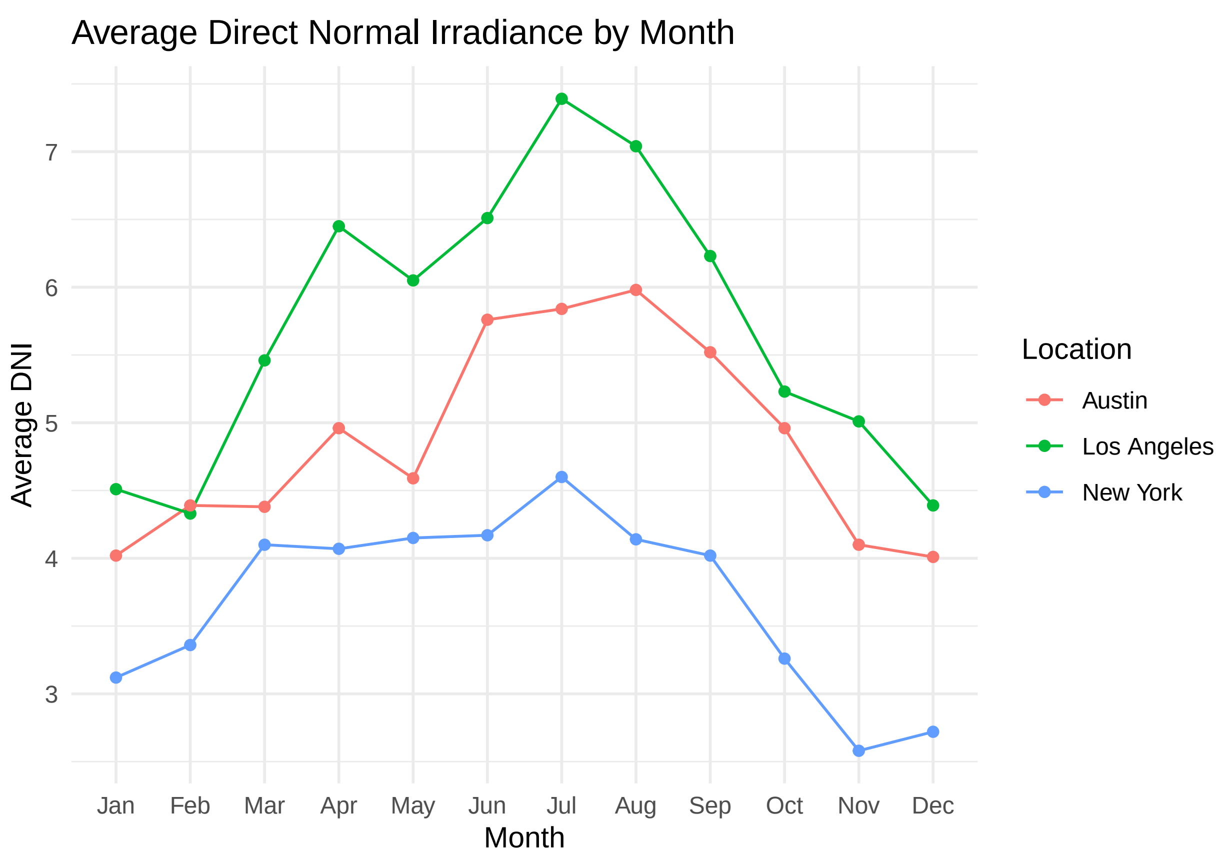
```r
# iterate over all locations, query the API for each, and
# combine into one dataframe
solar_resource_df <- locations %>%
    lapply(function(loc) {
        solar_api_json_get_df("api/solar/solar_resource/v1.json",
            queries = list(api_key = my_api, lat = loc$lat, lon = loc$lon)) %>%
            mutate(location = loc$name)
    }) %>%
    bind_rows()

solar_resource_df
```

```
##    month avg_dni avg_ghi avg_lat_tilt    location
## 1    Jan    3.12    1.97         3.55    New York
## 2    Feb    3.36    2.69         4.04    New York
## 3    Mar    4.10    3.86         4.86    New York
## 4    Apr    4.07    4.70         4.97    New York
## 5    May    4.15    5.45         5.18    New York
## 6    Jun    4.17    5.78         5.24    New York
## 7    Jul    4.60    5.98         5.58    New York
## 8    Aug    4.14    5.14         5.24    New York
## 9    Sep    4.02    4.23         5.00    New York
## 10   Oct    3.26    2.94         4.11    New York
## 11   Nov    2.58    1.99         3.26    New York
## 12   Dec    2.72    1.67         3.13    New York
## 13   Jan    4.51    3.06         4.88 Los Angeles
## 14   Feb    4.33    3.65         5.04 Los Angeles
## 15   Mar    5.46    5.15         6.19 Los Angeles
## 16   Apr    6.45    6.35         6.61 Los Angeles
## 17   May    6.05    6.89         6.51 Los Angeles
## 18   Jun    6.51    7.24         6.51 Los Angeles
## 19   Jul    7.39    7.65         7.06 Los Angeles
## 20   Aug    7.04    7.02         7.12 Los Angeles
## 21   Sep    6.23    5.79         6.68 Los Angeles
## 22   Oct    5.23    4.42         5.90 Los Angeles
## 23   Nov    5.01    3.46         5.39 Los Angeles
## 24   Dec    4.39    2.82         4.72 Los Angeles
## 25   Jan    4.02    2.87         4.18      Austin
## 26   Feb    4.39    3.54         4.61      Austin
## 27   Mar    4.38    4.25         4.83      Austin
## 28   Apr    4.96    5.45         5.53      Austin
## 29   May    4.59    5.90         5.50      Austin
## 30   Jun    5.76    6.63         5.93      Austin
## 31   Jul    5.84    6.61         6.05      Austin
## 32   Aug    5.98    6.34         6.29      Austin
## 33   Sep    5.52    5.23         5.81      Austin
## 34   Oct    4.96    4.11         5.18      Austin
## 35   Nov    4.10    3.16         4.45      Austin
## 36   Dec    4.01    2.71         4.11      Austin
```

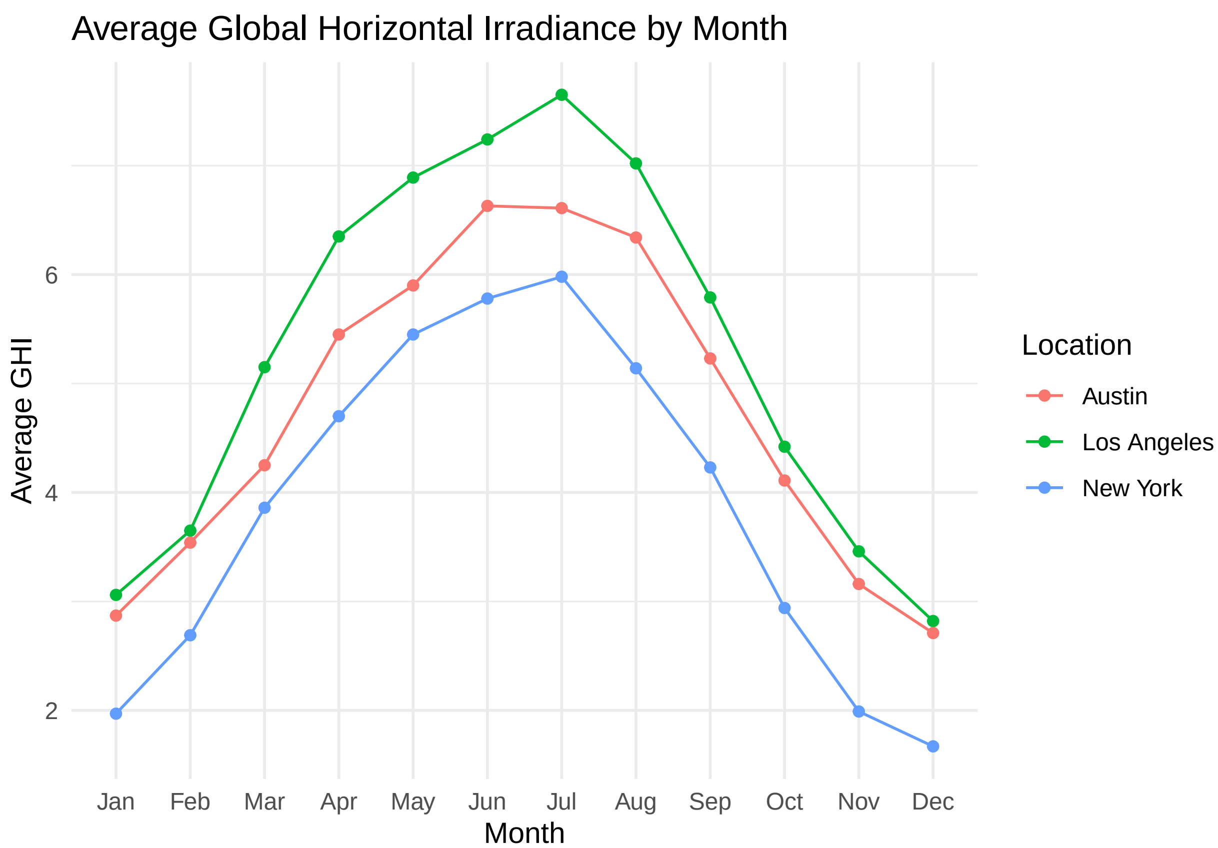
Visualising Solar Resource Data by Location

I analyse solar resource data over three dimensions: irradiance distribution, seasonality and financial return.

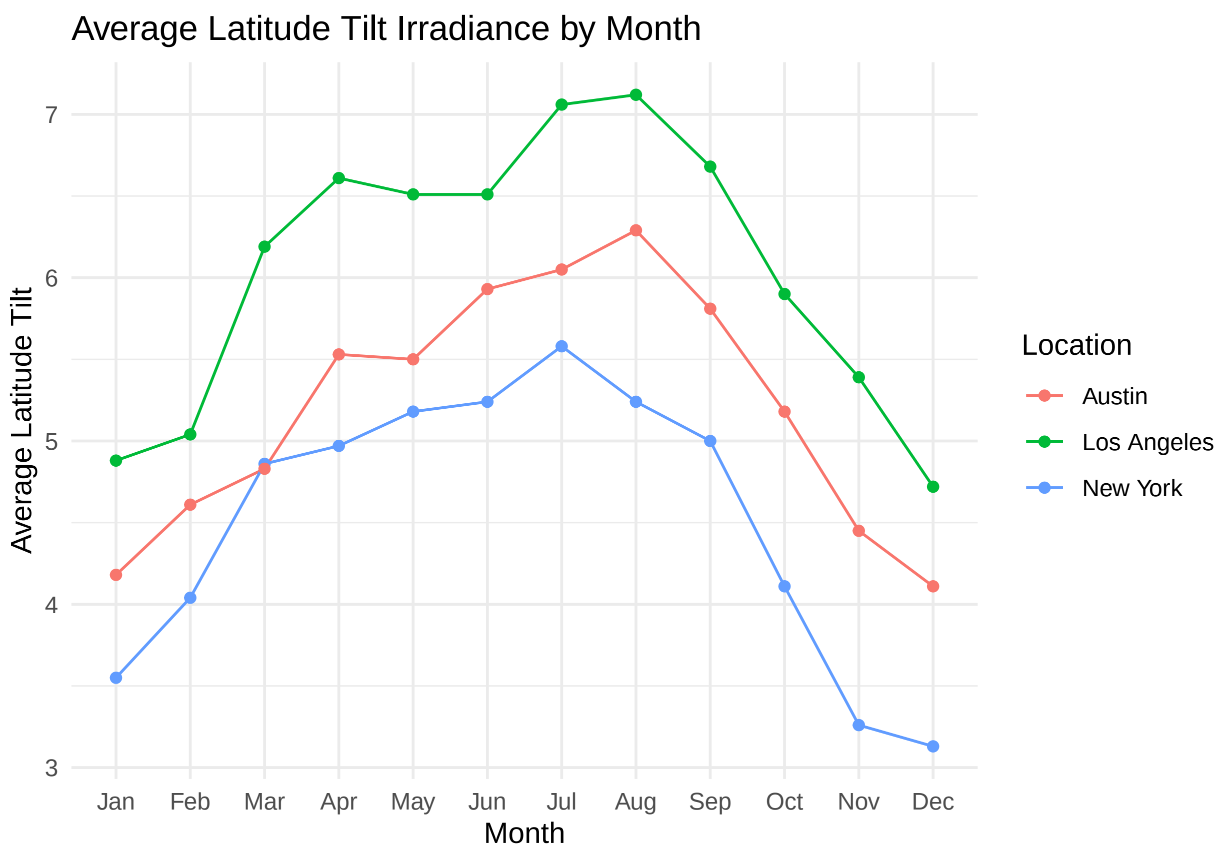
```r
# Average DNI by month, coloured by location Los Angeles
# shows the highest and most stable DNI year-round; New
# York and Austin show stronger seasonal variation.
# reusable plot function
plot_solar_metric <- function(data, metric, title, y_label) {
    data %>%
        mutate(month = factor(month, levels = month.abb)) %>%
        ggplot(aes(x = month, y = .data[[metric]], colour = location,
            group = location)) + geom_line() + geom_point() +
        labs(title = title, x = "Month", y = y_label, colour = "Location")
}

# one call per metric
plot_solar_metric(solar_resource_df, "avg_dni", "Average Direct Normal Irradiance by
Month",
    "Average DNI")
```




```r
plot_solar_metric(solar_resource_df, "avg_ghi", "Average Global Horizontal Irradiance by
Month",
    "Average GHI")
```



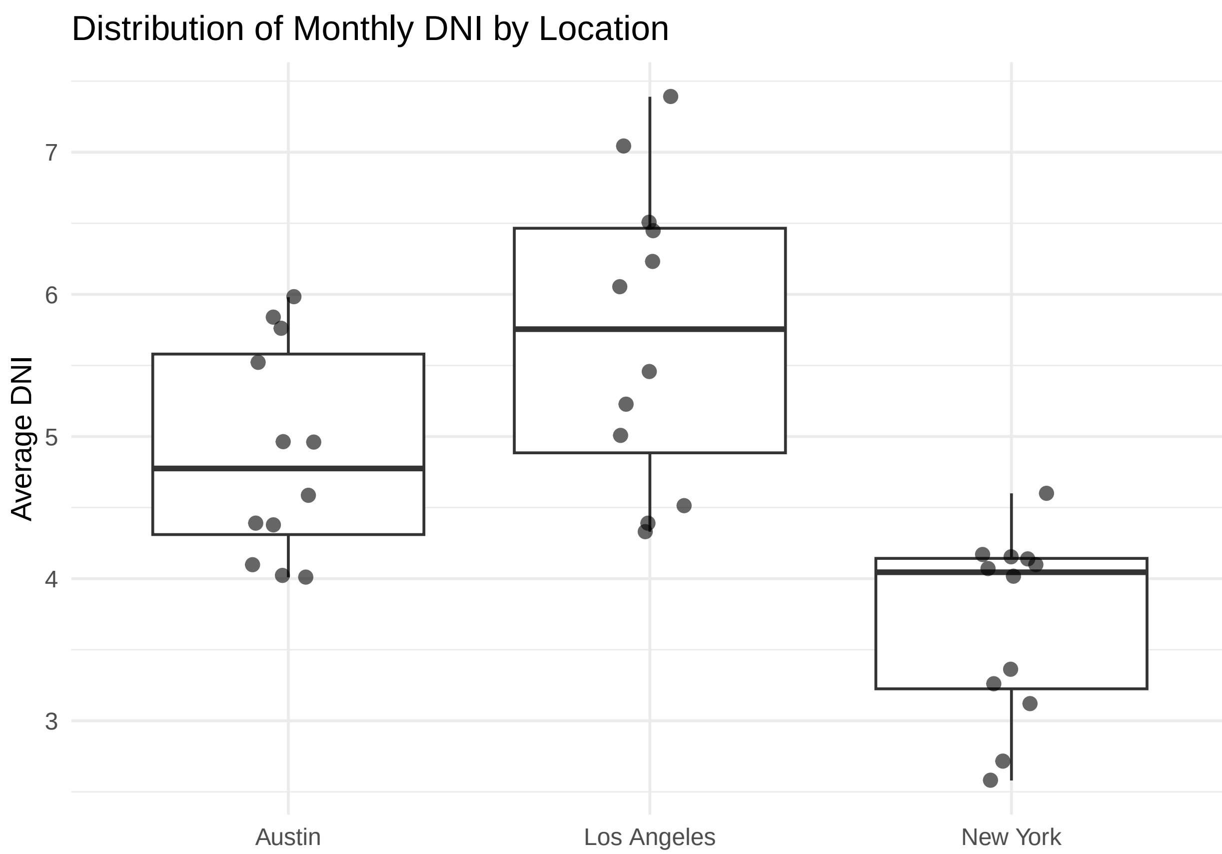

```
plot_solar_metric(solar_resource_df, "avg_lat_tilt", "Average Latitude Tilt Irradiance by
Month",
    "Average Latitude Tilt")
```



The plots show that Los Angeles has the highest and most consistent solar resource across all three metrics year-round. Austin shows strong summer peaks. New York has the most pronounced seasonal variation, with the lowest winter values of the three cities.

Solar Irradiance

```
solar_resource_df %>%
    ggplot(aes(x = location, y = avg_dni)) + geom_boxplot() +
    geom_jitter(width = 0.1, size = 2, alpha = 0.6) + labs(title = "Distribution of
    Monthly DNI by Location",
    x = NULL, y = "Average DNI")
```



The average DNI in New York is significnatly different from Los Angeles and Austin but has the small range and low mean. Therefore New York is least suitable for solar based on resource alone. However, as shown below, its high electricity price partially compensates.

Seasonality Ratio

How much worse is the worst month compared to the best?

```
solar_resource_df %>%
    group_by(location) %>%
    summarise(seasonality_ratio = min(avg_ghi)/max(avg_ghi))
```

```
## # A tibble: 3 x 2
##   location    seasonality_ratio
##   <chr>                   <dbl>
## 1 Austin                  0.409
## 2 Los Angeles             0.369
## 3 New York                0.279
```

The seasonality ratio in Austin (0.409) > Los Angeles (0.369) > New York (0.279). This suggests that battery storage would need to be largest in New York.

Energy Generation per Month

```r
# How much energy would be generated by month given a panel with 0.175
# efficiency solar_resource_df.
# Which city has the best return on investment for solar?

# costs per kwh from https://www.chooseenergy.com/electricity-rates-by-state/ (June 2026)
solar_resource_df %>%
  mutate(
    cost_per_kwh = case_when(
      location == "Austin"      ~ 16.39,
      location == "New York"    ~ 28.55,
      location == "Los Angeles" ~ 33.35
    ),
    # energy output: panel_size (kW) * avg_ghi * efficiency * days in month
    monthly_kwh     = 5 * avg_ghi * 0.18 * 30.4,
    # money saved by not buying that electricity from the grid
    monthly_savings = monthly_kwh * (cost_per_kwh / 100)  # convert cents to dollars
  ) %>%
  group_by(location) %>%
  summarise(
    annual_kwh     = sum(monthly_kwh),
    annual_savings = sum(monthly_savings)
  ) %>%
  arrange(desc(annual_savings))
```

```
## # A tibble: 3 x 3
##   location    annual_kwh annual_savings
##   <chr>            <dbl>          <dbl>
## 1 Los Angeles      1737.           579.
## 2 New York         1270.           362.
## 3 Austin           1554.           255.
```

```r
# How many months per year does each city exceed a viable solar threshold?
# (set as 4 kWh/m2/day)
solar_resource_df %>%
  group_by(location) %>%
  mutate(feasible = ifelse(avg_ghi > 4, "feasible", "not feasible")) %>%
  summarise(feasible_months = sum(avg_ghi > 4))
```

```
## # A tibble: 3 x 2
##   location    feasible_months
##   <chr>                 <int>
## 1 Austin                    8
## 2 Los Angeles               8
## 3 New York                  6
```

In terms of solar irradiance, Austin has the same number of feasible months as Los Angeles (8), and New York has fewer (6). However, in terms of electricity prices, the savings from using a rooftop solar panel in New York are greater than in Austin ($362 vs $255). Los Angeles performs best in terms of solar irradiance (average 1930 kwh/m2/year) and savings ($579).

Conclusion

This analysis compared rooftop solar potential across New York, Los Angeles, and Austin using irradiance data from the NLR API. Los Angeles has the strongest solar resource, generating an estimated 1,737 kWh/year from a 5kW panel, but all three cities exceed the 4 kWh/m^2/day viability threshold for at least six months of the year. The most counterintuitive finding is that New York, despite having the weakest solar irradiance and the fewest viable months, produces the second-highest annual savings ($362) due to its high electricity price of 28.55¢/kWh. Austin, despite matching Los Angeles on feasible months, produces the lowest savings ($255) because its electricity price is less than half that of the other two cities. These results suggest that electricity price is as important as solar resource when evaluating rooftop solar investment, and that New York residents in particular may be underestimating the financial case for solar adoption.